Predictor · normal → 90m / 6h / 10h

Three identical layers. One forward for each $t \in \{1, 2, 3\}$. The encoder stays frozen; these layers and $T[t]$ train.

Questions

$x \in \mathbb{R}^{B \times 129 \times 768}$

genes: $E[g] + R[r]$ or $m + R[r]$

last row: c

$+ T[t]$

$T[t] \in \mathbb{R}^{768}$, added to every row

Memory

$H_{\text{src}} \in \mathbb{R}^{B \times L_s \times 768}$

$L_s \le 512$

keys and values for cross-attention

pad positions drop out of the softmax

rebuilt at each t:
the 19 mask tokens move

× 3

$$\text{Attn}(Q, K, V) = \text{softmax}\left(QK^\top / \sqrt{96}\right)V$$

each block is $y = \text{LN}(x + \text{block}(x))$, then one more LN after layer 3

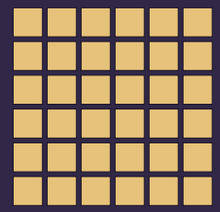
Self-attention

$Q = K = V = x$

8 heads, $d_k = d_v = 96$

$A \in \mathbb{R}^{B \times 8 \times 129 \times 129}$

every question sees all 129, including the cell question



129×129

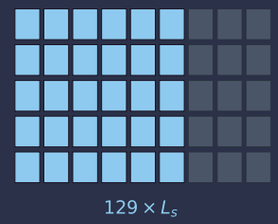
Cross-attention

$Q = x, \quad K = V = H_{\text{src}}$

$A \in \mathbb{R}^{B \times 8 \times 129 \times L_s}$

pad columns are masked

memory is this cell only, $L_s \le 512$



$129 \times L_s$

Position-wise FFN

GELU, the same map at each of the 129 positions

Leaves

final LN after layer 3

$\hat{z} \in \mathbb{R}^{B \times 129 \times 768}$

$\hat{z}_{\text{gene}}$

first 128 rows $\in \mathbb{R}^{B \times 128 \times 768}$

$\hat{z}_{\text{cell}}$

last row $\in \mathbb{R}^{B \times 768}$